

1. Find the area of the region enclosed by the curves $y = x^2$ and $y = 2x - x^2$.

2. Let R be the finite region bounded by the curves $y = 5 \tan(x)$, $y = 5$ and $x = 0$. Set up, but do not evaluate definite integrals which represent the area of the region R . Use your calculator to verify that both integrals produce the same value.
- (a) Integrate with respect to x .
- (b) Integrate with respect to y .

3. Let R be the finite region bounded by the curves $x = y^2$, $x = (y - 6)^2$, and $x = 0$. Set up, but do not evaluate definite integrals which represent the area of the region R . Use your calculator to verify that both integrals produce the same value.

(a) Integrate with respect to x .

(b) Integrate with respect to y .

4. Let R be the finite region bounded by the curves $y = 2\ln(x)$, $y = 3\ln(x)$, and $y = 6$.

(a) Determine the area of R by evaluating a definite integral with respect to y .

- (b) Set up, but do not evaluate, definite integral to represent the area of R by evaluating by integrating with respect to x . Use your calculator to see that this integral is equivalent to the one above.

5. Let R be the finite region bounded by the curves $f(x) = -2x + 14$ & $g(x) = \frac{20}{x}$.

- (a) Determine the area of R by integrating with respect to x .

(b) Determine the area of R by integrating with respect to y .

6. Find the of the region bounded by the curves $y = \frac{x}{\sqrt{x^2 + 1}}$ and $y = x^4 - x$.

(Use your calculator to help find points of intersection and round decimals to 2 places)